

**High Resolution fMRI Reveals Configural Processing
of Cars in right anterior Fusiform Face Area of Car Experts**

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Abstract

Visual object expertise correlates with neural selectivity in the fusiform face area (FFA). While behavioral studies suggest that visual expertise is associated with increased use of holistic and configural information, little is known about the nature of the supporting neural representations. Using high-resolution 7T fMRI we recorded the multi-voxel activation patterns elicited by whole cars, configurally-disrupted cars, and car parts in individuals with a wide range of car expertise. A probabilistic support vector machine (SVM) classifier was trained to differentiate activation patterns elicited by whole car images from activation patterns elicited by misconfigured car images. The classifier was then used to classify new combined activation patterns that were created by averaging activation patterns elicited by individually-presented top and bottom car parts. In line with the idea that the configuration of parts is critical to expert visual perception, car expertise was negatively associated with the probability of a combined activation pattern being classified as a whole car in the anterior right anterior FFA (rFFA2), a region critical to vision for categories of expertise. Thus, just as found for faces in normal observers, the neural representation of cars in rFFA2 is more holistic for car experts than car novices, consistent with common mechanisms of neural selectivity for faces and other objects of expertise in this area.

Expertise identifying visually similar objects is associated with category-selective responses in the Fusiform Face Area (FFA; e.g., Gauthier et al., 2000). Correlations between behavioral performance in visual tasks and FFA selectivity have been shown for many categories, such as faces (McGugin et al., 2017; Elbich & Scherf, 2017), cars (McGugin, Gatenby et al., 2012), birds (Gauthier et al., 2000), radiographs (Bilalic et al., 2014) and chess configurations (Bilalic et al., 2011)¹.

While visual expertise is often associated with category-selective responses in FFA, category-selective responses do not guarantee good performance with an object category. For example, individuals with developmental prosopagnosia, who by definition have seriously impaired face recognition, have seemingly normal face-selective responses in FFA (Avidan, 2014; Furl et al., 2011). Addressing this apparent contradiction, a study using multivariate pattern analysis (MVPA) revealed that the FFA response to faces in these patients is not normal (Zhang et al., 2015). In typical individuals, the response pattern across FFA voxels can distinguish between a normal face and a misconfigured face, in which the top and bottom parts have been switched. Interestingly, for individuals with developmental prosopagnosia, this gross configural change does not affect the response pattern in FFA (see also Schiltz et al., 2010; 2006). This finding is consistent with reports of abnormal configural and holistic processing in individuals with prosopagnosia (Liu & Behrmann, 2014; Marotta et al., 2002) and suggests that the sensitivity of FFA response patterns to configuration may predict expert performance better than mere selectivity.

¹ Where expertise effects have not been observed (e.g., Grill-Spector et al., 2004; Harel et al., 2010), proposed mediating factors include the specificity of the stimuli (e.g., modern car experts tested with antique cars, Bukach et al., 2006; Bukach et al., 2010) and competition between categories of expertise and attentional factors (e.g., Harel et al., 2010; McGugin, Van Gulick et al., 2014).

The question therefore arises: In a non-face domain of expertise, would the response of the FFA depend on configuration or would it be selective regardless of configuration as it is for faces with prosopagnosic patients? If responses to non-face objects in FFA arise for reasons similar to the face responses, then they should be equally dependent on configuration. Specifically, here, we predict that expertise with cars should be associated with sensitivity to the configuration of cars parts in FFA, whereas car novices would show less such sensitivity; while FFA representations may not have any sensitivity to configuration in prosopagnosic patients, we would not go as far as equating being a car novice to having a face recognition impairment. There is some indication that perceptual expertise can increase configural processing. Training with novel objects produces holistic processing, demonstrated by a failure to selectively attend among the available object parts despite task demands to do so, effects that are sensitive to configuration (Gauthier et al., 1998; Gauthier & Tarr, 2002; Chua et al., 2015; Wong, Palmeri & Gauthier, 2009). Moreover, car experts process cars holistically only when car parts are in a normal configuration (Gauthier et al., 2003; Bukach et al., 2010). Studies have reported correlations between fusiform gyrus selectivity for trained objects and holistic processing of these objects in their trained configuration (Gauthier & Tarr, 2002; Wong, Palmeri & Gauthier, 2009). Yet no direct measurement of sensitivity to configuration in neural responses evoked by non-face objects in real-world expertise has been conducted.

Here we ask whether car expertise predicts how well multi-voxel response patterns in FFA discriminate configural changes in car parts. We first sought to replicate standard expertise effects, measuring car-selectivity in posterior and anterior FFA (FFA1 and FFA2, respectively), Occipital Face Area (OFA) and Lateral Occipital Complex (LO), expecting car expertise to correlate with car selectivity. Note that in previous work, car expertise has predicted selective

responses to cars in both face-selective and in non-face-selective areas (McGugin et al., 2012; Harel et al., 2010), but effects in many of these areas disappeared with increased attentional and perceptual loads, leaving a robust correlation between car expertise and neural selectivity only in the right FFA2 (McGugin et al., 2014a). Other studies found stronger effects of experience or expertise in FFA2 (Golarai et al. 2017, McGugin et al., 2014b; 2017). Accordingly, we expect that most of our ROIs may show an effect of car expertise (as measured here in a low-demand 1-back task) but that the right FFA2 may be the main region that is sensitive to configuration of car parts as a function of car expertise.

To test this hypothesis, we measured BOLD fMRI activity patterns while subjects viewed images of intact cars, misconfigured cars (location of the top and bottom parts exchanged), and individually presented top and bottom car parts. To assess whether car representations were sensitive to configuration, we trained a support vector machine (SVM) classifier to differentiate the multi-voxel patterns elicited by whole and configurally-disrupted cars in each ROI. For each subject, we asked the SVM to quantify the similarity of a combined linear combination of the voxel pattern elicited by separate top and bottom car parts to either the pattern elicited by whole cars or to the pattern elicited by configurally-disrupted cars. Based on prior work (MacEvoy & Epstein, 2009), we used an equal combination of the patterns elicited by each part. We expect that in car novices, cars would be represented in a part-based manner and so a combination of parts would be roughly equally similar to whole cars and to misconfigured cars. Prior work shows that expertise effects with cars are reduced when the configuration of halves is disrupted (e.g., Gauthier et al., 2003; Bukach et al., 2010). Accordingly, we expect that car representations should become increasingly sensitive to configuration as a function of car expertise, and that the combined pattern, which cannot carry information about whole car configuration, should look

less like the patterns elicited by whole cars and consequently more like the patterns elicited by misconfigured cars.

Methods

Subjects

Thirty adult males with normal or corrected-to-normal vision were recruited for this experiment, with two subjects excluded from further analyses for failure to comply with instructions during behavioral testing (remaining subjects, $M=24.6$ years, $SD=5.5$). A deliberate effort was made to recruit subjects with a broad range of car expertise, although all our subjects, even novices, are expected to have some experience with car models. This was effective in increasing variability especially at the high end of the spectrum of ability. In a sample of 213 subjects not recruited for expertise with any category (VanGulick et al., 2015), the mean and SD for the VETcar (see description below) was .60 (.14). The present sample has a mean of .69 (.21), with no subject falling 2 SD below the mean but 6 subjects (20%) falling 2 SD above the mean. The sample size was based on an expected effect size of $r = .5$ for car expertise correlations with FFA activity (McGugin, Van Gulick et al., 2014), which requires a sample size of 26 subjects for 80% power at an alpha of .05. Subjects participated in the study for monetary compensation, and written informed consent was obtained in accordance with a protocol approved by the Vanderbilt Institutional Review Board.

Behavioral Tests and Stimuli

All subjects completed a Sequential Matching Task (e.g., Gauthier et al. 2000), a Vanderbilt Expertise Test battery (VET; McGugin, Richler, et al. 2012; McGugin, Van Gulick et al., 2014), and the Cambridge Face Memory Test (CFMT; Duchaine & Nakayama, 2006).

The sequential matching task comprised 1 s, with objects from each of three categories presented in separate block: birds, cars or planes. On each trial, subjects judged if two sequentially presented images showed the same object (same make and model of car or airplane, or same species of bird) or not; images were always different. The first stimulus of each pair was displayed for 1000ms, followed by a 500ms mask. The second stimulus was then displayed until a response was made (up to a maximum of 5000ms).

The VET battery included 8 separate tests for 8 different categories: leaves, owls, butterflies, wading birds, mushrooms, cars, planes, and motorcycles (see McGugin, Richler, et al. 2012, for details). For each category, subjects first studied a set of 6 images, with each image showing a different species/model from the test category. This was followed by 48 3-alternative forced choice trials comprising a target image of one of the 6 studied species/models and two foil images of unstudied species/models from the from the test category.. For the first 12 trials of each block, the target image was an exact match to one of the 6 studied images, while for the remaining 36 trials it was a new image matching the identity of one of the 6 studied species/models (new target/foil images were used for each test trial) .

In the CFMT (Duchaine & Nakayama, 2006; Russell, Duchaine & Nakayama, 2009), subjects studied frontal views of 6 target faces, followed by a short learning phase in which the subjects saw cartoon faces to familiarize them with the task. Subjects were then presented with 54 3-alternative forced choice test trials containing one target face and two distractor faces.

Matching faces varied in lighting, pose, or both relative to their studied presentation. In addition, the final 24 trials were presented with faces embedded in Gaussian noise to increase difficulty level.

Structural and Function Magnetic Resonance Imaging (MRI)

Imaging was performed using a Philips Medical Systems 7-Tesla (7T) human MRI scanner located in the Vanderbilt University Institute of Imaging Science. All scanning was performed using a 32-channel SENSE parallel-imaging head coil. Functional scanning used echo-planar imaging (EPI) protocols. All subjects completed a single scanning session consisting of a structural scan, a standard-resolution localizer, a high-resolution localizer and 6-8 high-resolution experimental runs. We used the IView Bold analysis package provided by Philips to place the high-resolution slices for each subject so that they included the FFA1 and FFA2 identified during the standard resolution scan.

Structural scan: Structural T1 scans were acquired using 249, 0.7 mm sagittal slices with a 352 x 351 matrix covering a field of view of 246 x 246 mm, for a final isotropic resolution of 0.7 mm. Structural scans used a TR of 4.761 ms, a flip angle of 7 degrees, and a TE of 2.1 ms.

Standard-resolution functional localizer: Standard-resolution functional scans covering nearly the whole brain were acquired to define our functional ROIs. These scans used a 3D PRESTO sequence covering a volume of 211 x 211 x 85 mm. Functional volumes were reconstructed into 34, 2.5 mm axial-oblique slices with a 96 x 96 matrix for a final in-plane resolution of 2.198 x 2.198 mm. Standard-resolution scans used a volume acquisition time of 2 s, a TR of 15.16 ms, a

flip angle of 12 degrees, and TE of 20.06 ms. Following preprocessing, standard-resolution scans were resampled to an isotropic resolution of 3 mm. Standard-resolution localizer scans consisted of 256 volumes (512 s). In addition to the standard processing pipeline, un-resampled standard-resolution functional localizer scans were analyzed in real time using IView to aid in placement of high-resolution slices (see below).

During the localizer, subjects completed a one-back repetition task while viewing blocks of faces, objects, body parts and scrambled objects. Each block comprised 16 images from one of the four categories with a total of 8 blocks of each category (4 categories x 8 blocks). Stimuli were drawn from a pool of grayscale photographs of 36 male and female Caucasian adult faces, 36 images of common household objects, 206 body-part images – including whole bodies (without heads), individual limbs, hands and feet – and 40 scrambled images – created by dividing a separate set of images of airplanes into squares and scrambling their locations. Each stimulus appeared on screen for 1 s and no two consecutive stimuli were the same size. Repeats occurred 1-2 times per block and subjects were instructed to press a button only for immediate repetitions.

High-resolution functional localizer: High-resolution functional scans were acquired to verify category selectivity and effects of expertise in functional ROIs. These scans used echo-planar imaging with 32 contiguous 1.5 mm axial-oblique slices placed to cover regions initially localized using the standard-resolution functional scans with real-time analysis. Volume acquisition time (TR) was 4 s, flip angle was 65 degrees, and TE was 25 ms. These scans were acquired with a 148 x 145 matrix covering a field of view of 224 x 224 mm, and were reconstructed using a 160 x 160 matrix for a final in-plane acquisition of 1.4 x 1.4 mm.

Following preprocessing, high-resolution scans were resampled to an isotropic resolution of 1 mm. High-resolution localizer scans consisted of 80 volumes (320 s). Because of the limited brain coverage necessitated by high-resolution imaging and the individualized slice placement to target ROIs identified in IView during the standard-resolution localizer, high-resolution coverage varied across subjects. In order for coverage to include the targeted ROIs, we placed the slice stack such that it missed the anterior temporal poles, the occipital pole and the frontal pole in most subjects.

During the high-resolution localizer, subjects completed a one-back repetition task while viewing blocks of faces, objects, cars and scrambled objects. Each block comprised 16 images from one of the four categories with a total of 5 blocks of each category (4 categories x 5 blocks). Stimuli were drawn from a pool of 40 grayscale photographs of Caucasian male faces as well as 70 car images. The car images were from modern car models and did not overlap with the images used in the experimental task. The stimulus presentation and task instructions were the same as in the standard-resolution localizer.

High-resolution functional experimental task: Experimental-task scans were acquired to test our predictions about configural sensitivity using multivoxel pattern analysis. The slice placement and scanning parameters were the same as for the high-resolution functional localizer scans, except that each task scan consisted of 50 volumes (200s). In each experimental run, subjects completed a motion-detection task comprising 8 blocks of 20 images. The images in each block came from one of four conditions - whole car images, misconfigured car images, isolated top car-part images, or isolated bottom car-part images - with two blocks of each stimulus condition per run. Task blocks were interleaved between 8 s fixation blocks with each run beginning and

ending with fixation. Each task block contained 20 images, each presented for 600 ms and followed by a 200 ms blank interval. For the duration of each run, subjects were instructed to fixate on a dot in the center of the screen. To engage their attention, a randomly selected 4 images in each block moved 5 pixels to the left or to the right during stimulus presentation. Subjects were asked to indicate the direction of these movements by button press. For the intact and the configurally-disrupted images, the movement would occur either on the upper or the lower part to encourage the subjects to attend equally to both parts of the images. All but three of the subjects completed 8 blocks of the task with two completing 7 blocks and one completing 6 blocks.

The car images were grayscale and showed a three-quarter front view of a car facing to the right. Each car image was divided into two parts, with a box surrounding each part. The top part contained the roof and windows and the bottom part contained the wheels and most of the body of the car. Each car image was used to create one of the four stimulus types (see Figure 1): whole cars, in which the top and bottom were presented in the correct configuration; misconfigured cars, in which the top and bottom halves were switched; and isolated top and bottom car-halves. The size of the upper and lower car parts on the screen were $1.9^\circ \times 14.5^\circ$ and $2.9^\circ \times 14.5^\circ$, respectively. The intact and misconfigured car images were simply the vertical combination of the two parts (i.e. $4.8^\circ \times 14.5^\circ$). All images were presented centered on a fixation dot located in the center of the screen.

Because our goal was to predict decoding as a function of car expertise rather than to decode location, we presented car parts at fovea rather than in their original positions in whole images. We do not expect that the positioning of these parts could drive SVM decoding because the population receptive fields of voxels in FFA have a median size of 3.4° when stimuli are

attended (Kay et al., 2015), and our parts were displaced a maximum of 1.5 degrees. In addition, the parts overlapped by 50% in their location across the whole and part conditions, further suggesting they should engage largely overlapping cell populations. Finally, we focus here on the relative effects across subjects, not expected to interact with receptive field effects.

Data Analyses

Behavioral data: For each subject, d' scores were calculated for each category in the Sequential Matching Task (bird, car, and plane) and percent correct scores were calculated on the CFMT and for each category on the VET (butterflies, cars, leaves, motorcycles, mushrooms, owls, planes, and wading birds). Aggregate car task scores were computed as the mean of the z -transformed scores for cars on the Sequential Matching Task and VET. To obtain an aggregate measure of car-specific expertise, we used the non-vehicle categories of the Matching Task and VET (birds, leaves, owls, butterflies, wading birds, and mushrooms) as a baseline, computed non-vehicle aggregate scores, and regressed them out of the car aggregate scores². We used only the non-vehicle categories to control for general object recognition because, as in McGugin, Richler, et al. (2012), performance with cars was highly correlated with performance on the other two vehicle categories (planes and motorcycles); this is likely because of joint experience and interest. For the Sequential Matching Task, performance on car trials was correlated with performance on plane trials, $r=.67$, $p=.0001$, and for the VET, performance on car trials was correlated with performance on motorcycle trials, $r=.63$, $p=.0003$, and marginally significantly with performance on plane trials, $r=.37$, $p=.0542$. In the remainder of this paper, we simply refer to residualized car aggregate scores as “car expertise.”

² Two subjects had very low scores on one of eight the VET categories (z -scores -2.5 and -3). For these subjects, we omitted these categories when calculating the non-vehicle aggregate, however the choice to omit these categories did not change any of the reported findings.

Functional data: The functional data were analyzed using BrainVoyager, in-house MATLAB code written using the NeuroElf toolbox (neuroelf.net), and LibSVM (Chang & Lin, 2011). High-resolution T1 scans were used for registering the functional scans and calculating Talairach warping parameters. Functional scan preprocessing in BrainVoyager included slice acquisition time correction (high-resolution EPI scans only), 3D motion correction, linear trend removal, and temporal high-pass filtering (3 cycles per run). The data from all high-resolution runs were interpolated to 1 mm isotropic space and warped to Talairach space (Talairach & Tournoux, 1988). Standard- and high-resolution localizer runs (used in the univariate analyses) were spatially smoothed, which is considered optimal for detection purposes (Worsley et al., 1992), with a 6 mm Gaussian kernel but experimental-task runs (used in the multivariate analyses) were left unsmoothed.

Standard-resolution functional localizer: To identify FFA1, FFA2, OFA, and LO we ran separate GLMs for each subject, with regressors for faces, body parts, objects, and scrambled objects, on the standard-resolution localizer runs (Friston et al., 1995). The individual-subject centers of FFA1, FFA2, and OFA were defined by the peaks of the contrast faces minus objects (t-test, $p < .05$) in middle-fusiform (posterior: FFA1, anterior: FFA2) and inferior-occipital cortex, respectively. The center of LO was defined by the peak activation of the contrast objects minus scrambled objects in the lateral occipital cortex (t-test, $p < .05$). An additional, body-parts minus objects contrast was used to help distinguish FFA1 and FFA2, which are typically separated by a body-part selective area (Weiner & Grill-Spector, 2010). We aimed to create ROIs that were equal in size (to yield comparable sensitivity for MVPA) and that were sufficiently large for

MVPA (Kriegeskorte et al., 2006). ROI size varies slightly because voxels that were initially included in more than one ROI were excluded from all ROIs, as were voxels with a raw BOLD signal value of less than 20, which excludes voxels that clearly fall outside the brain. To standardize the spatial extent of our ROIs across subjects, we defined ROIs as spheres (radius 7.5 mm) centered on the peak t-value for each contrast. Talairach coordinates, size and number of individuals are reported in Table 1. Final ROI size was not correlated with car expertise ($r_s < .15$, $p_s > .49$).

High-resolution functional localizer: The high-resolution functional localizer runs were similarly subjected to a GLM analysis (with regressors for face, car, object & scrambled object blocks) to extract parameter (beta) weights and measure the relationship between car expertise and univariate face and car-selectivity in each individual-subject ROI.

High-resolution experimental task: The data from the experimental-task runs were analyzed using a linear probabilistic SVM trained and tested on the multi-voxel activation patterns within each ROI (Chang & Lin, 2011). For each ROI, the pattern of BOLD activation across voxels at each timepoint was extracted and z-scored. For each 16 s block (i.e., 4 TRs), a mean activation pattern was computed for the 16 s period beginning 4 s after the block onset and ending 20 s after block onset (i.e. offset forward in time by 1 TR). We first trained and then tested a SVM to estimate the probability that a given activation pattern belonged to a whole car block vs. a misconfigured car block, using leave-one-block out cross validation to ensure independence of training and test data. That is, the SVM yielded a single probability for each test pattern representing the evidence that the pattern belonged to a whole car block. Note, because the

choice of target category is not important (i.e., a .8 probability that a test pattern belonged to a whole car block implies a .2 probability that it belonged to a misconfigured car block), we combined the SVM probabilities for whole and misconfigured car-pattern classification by taking 1 minus the classification probability on misconfigured car-pattern test trials and calculating the mean probability correct (Table 4). Next, we sought to investigate if the combined activation patterns (averaged top-only and bottom-only activation patterns) were equally similar to activation patterns elicited by whole and misconfigured cars (consistent with parts-based representation), or were more similar to activation patterns elicited by whole cars than misconfigured cars (consistent with holistic representation). Thus, we trained the probabilistic SVM using the full set of activation patterns from whole and configurally-disrupted car blocks and tested it on combined activation patterns, using the probability estimates as an index of similarity. Combined activation patterns were created by exhaustively pairing the 16 activation patterns from the top and bottom car-part blocks and computing the mean activation in each voxel.

Results

Univariate Analyses: Expertise Effects in Car-Selectivity

First, we sought to replicate prior work demonstrating expertise effects in ventral occipitotemporal cortex (Gauthier et al., 2000; McGugin, Gatenby et al., 2012; McGugin, Newton et al., 2014; McGugin, Van Gulick et al., 2014; Xu 2005). To maximize our power, given that several studies have reported expertise effects in these ROIs and since we have a clear directional prediction, we report one-tailed tests for positive correlations. We do not correct for multiple comparisons and invite readers to treat r-values as estimates of effect size. In the ROIs

defined in the standard-resolution localizer, we measured the univariate response to cars (beta weights) relative to the average for the non-car conditions (Faces, scrambled objects, objects) in the high-resolution localizer. Because most fMRI studies of car expertise have subtracted neural activity for non-car conditions from the neural response to cars, we report correlations of car expertise with a subtracted score. Because partialing out activity for non-cars from the response to cars is arguably a better approach (so that the non-car variance does not contribute to the correlation with behavior), we also report partial correlations between car expertise and car residuals (neural activity for cars, regressing out neural activity for non-cars). The two methods are generally consistent and show significant or near significant car expertise effects in most areas, with the exception of left OFA and right FFA1 (see Table 2). Because we also collected CFMT scores, we report similar analyses for faces, relating CFMT performance to either neural activity for faces either subtracting the three non-face conditions, or partialing them out. As in other work (McGugin et al, 2017), face recognition behavior does not show robust correlations with neural selectivity for faces in most ROIs. This may be due to the greater variation in both behavior and neural activity for cars than faces in our sample of subjects recruited to vary in car expertise (e.g., coefficient of variation for CFMT: 17.7%, for VET-car: 30.4%; for beta weights for faces in rFFA2: 82%, for beta weights for cars in rFFA2: 132%).

Univariate Analysis: Sensitivity to configuration

The mean univariate responses to whole cars and misconfigured car images in each ROI are shown in Table 3. Note that misconfigured cars still have easily recognizable features within each half (while misconfigured, they are not scrambled). Only a subset of the ROIs where car expertise predicted neural activity for cars showed significantly greater responses for whole cars

than for misconfigured cars: the left and right FFA2 and the right LO. These effects may represent a non-specific reduction of activity for misconfigured objects as they were generally not related to car expertise, except in the right LO, where the response to whole cars (partialing out the response to misconfigured cars) was moderately associated with car expertise ($r=.43$). Because these analyses were exploratory, we only report confidence intervals around effect sizes.

Multivariate Analyses: Configural Sensitivity

Our primary goal was to investigate whether the FFA would show patterns of responses to car parts that were more similar to the response to misconfigured cars than the response to whole cars, as a function of car expertise. To this end, we first trained a probabilistic SVM (Chang & Lin, 2011; McGugin, Newton et al., 2014) to distinguish activation patterns elicited by configurally-intact whole car images from those elicited by misconfigured car images. Testing the classifier with leave-one-block-out cross validation revealed above chance ($p<.05$) classification in all ROIs (Table 4). Just as for the corresponding univariate analysis, we did not have predictions for which ROIs should be better able to classify whole vs. misconfigured cars, a fairly easy categorization that all ROIs can perform on average. Because these analyses were exploratory, we only report confidence intervals around effect sizes. Effect sizes were relatively large in the right OFA, FFA1 and FFA2.

To test our main hypothesis, we then used the classifier to classify combined patterns (linear combinations of the voxel patterns elicited by top and bottom car parts when presented in isolation). If car-expertise is related to sensitivity to configural information, then we would expect that in subjects with greater expertise, the classifier would tend to classify the pattern of activity elicited by the combined patterns as more similar to misconfigured cars and less similar

to whole cars. The predicted correlation between expertise and classification of combined patterns as misconfigured was significant in right FFA2 (Table 4, Figure 2).

Discussion

For the first time, we demonstrate an association between object expertise and configural information represented by multi-voxel BOLD activation patterns in right FFA2. As a function of car expertise, representations in this region exhibit sensitivity to configuration – a hallmark of holistic representation – as shown by a classifier’s tendency to judge a combined pattern of car parts as less car-like. These findings help to link previous fMRI work demonstrating an increase in univariate BOLD selectivity for objects of expertise with behavioral studies that suggest experts make greater use of configural information than novices (Gauthier & Tarr, 2002; Chua et al., 2015; Wong, Palmeri, & Gauthier, 2009; Wong, Palmeri, Rogers et al., 2009; Bukach et al., 2010).

In addition, we replicated the previously reported increase in univariate BOLD selectivity to cars relative to other objects in several face-selective areas. As we found previously (McGugin et al., 2014), these effects are distributed among several face-selective regions as well as non-face-selective areas (LO). We demonstrated previously that, of these regions, the right FFA2 is most robust in its relation to experience with cars (McGugin et al., 2014) and with faces (McGugin et al., 2017), showing an effect of expertise even under manipulations that abolish the effects in other ROIs. The anterior FFA2 is closer than the FFA1 to an area reported by Nestor et al. (2011), which is not face-selective but codes for face identity, and to even more anterior areas such as the perirhinal cortex, which are important to the representations of complex conjunctions of features (O’Neil et al., 2013). These regions, especially in the right hemisphere, emerge as a

network that is important in the creation of more abstract representations of faces, objects or scenes (Barense et al., 2010).

Consistent with our results, a number of recent studies suggested that univariate BOLD selectivity, the original index of selectivity for faces and a standard measure in fMRI studies of expertise, may be too liberal an index of neural selectivity to relate to behavior. For example, univariate BOLD activation to faces is not always sufficient to differentiate developmental prosopagnosics (DPs) – a group defined by their below average face expertise – from controls (Avidan, 2014; Furl et al., 2011; Zhang et al., 2015). In other cases, the correlation between behavioral face expertise and univariate BOLD has been inconsistent (e.g. Jiang et al., 2013; Peirce et al., 2004), although as we have discussed, this may sometimes be a function of limited variability in face expertise. In contrast, multivariate measures have shown promise in relating behavioral measures to neural representations in FFA, even in cases where there was no relationship between univariate BOLD and expertise (e.g., Zhang et al., 2015; Jiang et al., 2013; McGugin, Van Gulick et al., 2014).

It is not yet clear why MVPA measures should index expertise effects more reliably than univariate BOLD activation. In some cases, it may be that MVPA simply provides a more sensitive measure (Jiang et al., 2013). In other cases, the results are more qualitatively different (e.g., Zhang et al., 2015; McGugin, Van Gulick et al., 2014), suggesting that there may be substantial heterogeneity of responses across voxels within an ROI (although this variability may not necessarily indicate a multidimensional code (Davis et al., 2014). The approach that we took here was specifically intended to measure the amount of configural information present in the multi-voxel pattern of activity extracted from each of our ROIs. A similar measure has been used in the face recognition literature to differentiate DPs and controls even in the absence of any

differences in univariate BOLD activation (Zhang et al., 2015), in line with behavioral evidence that DPs have deficits in configural processing relative to controls (Liu & Behrmann, 2014; Marotta et al., 2002). We should emphasize, again, that car novices are in many ways not comparable to prosopagnosic patients: the contrast between car novices and car experts is more likely quantitative (and indeed, we treat expertise as a continuum) whereas there are more striking differences between DPs and controls. For instance, congenital prosopagnosia is considered a disconnection syndrome (Avidan et al., 2013) whereas experts in a domain would not be expected to have qualitatively different connectivity between areas compared to novices. Nonetheless, our expectation that the FFA's sensitivity to car configuration as measured via MVPA may be a more stringent test of car expertise is based on these neuropsychological findings, which suggest a general framework in which to understand brain-behavioral relations for patients and controls, for both faces and objects. In addition, our operationalization of a configural representation has a good face validity as it directly relates to behavioral studies that have used similar stimuli and found behavioral effects of expertise (Gauthier & Tarr, 2002; Chua et al., 2015; Wong, Palmeri, & Gauthier, 2009; Wong, Palmeri, Rogers et al., 2009; Bukach et al., 2010).

In addition to testing the standard correlation between car-selectivity and expertise and our main prediction for the combined representations of car parts, we also conducted exploratory analyses of whether car expertise predicted the univariate activity for whole cars (controlling for the response to scrambled cars) and the classifier's ability to decode whole from scrambled faces. We had no clear predictions for these comparisons since they rely on contrasts that should be easily processed even by novices, but we note that in addition to showing significant or near significant effects in the FFA, these analyses revealed expertise effects in early areas (the right

OFA for MVPA and the right LO for the univariate effect). While these effects could be due to feedback projections from FFA, prior work suggests the right OFA is the first area differentiating faces from objects and that it is important in representing parts (Pitcher et al., 2007). It may play a similar role for other categories of expertise and some work suggests the representation of parts changes in expertise (Chua et al., 2015). Finally, while it may seem surprising that the domain-general area LOC shows an effect of car expertise, this could also represent tuning with experience of mechanisms that are engaged regardless of expertise. Indeed, this area and the right and left FFA2 showed a larger response to whole than misconfigured cars in all subjects, consistent with these regions known sensitivity to scrambling (Grill-Spector et al., 2000). An interesting and speculative possibility is that the univariate measure is more sensitive than the MVPA to attentional effects, with greater attention increasing the BOLD activation. In support of this suggestion, McGugin et al. (2014) reported that correlations between car expertise and mean BOLD selectivity were abolished within FFA and OFA when attention was directed towards a different object (a butterfly) whereas correlations between car expertise and the information content of multi-voxel patterns was not affected by redirecting attention (McGugin, Van Gulick et al., 2014). Indeed, while behavioral expertise typically requires considerable experience, there is evidence that online task-demands can increase the univariate BOLD response to non-expert object categories (Haist et al., 2010) and faces from an artificially created in-group (Van Bavel et al., 2011).

In the approach we used here, one consideration is how the response to individual parts should be combined in our combined representations. We used an equal combination of the patterns elicited by a particular part combination, as has been used in some previous work (e.g., MacEvoy & Epstein, 2009). Our results did not depend on this choice, as we found a similar

pattern of results when top and bottom parts were weighted 25/75 or 75/25. Baeck et al. (2013) compared different methods and found that the response to a pair of objects was best predicted by a weighted average of the responses to each object in the pair with a larger weight for the object that elicits a maximal response on its own, although the average response was nearly as effective. Interestingly, when the task required attending to a meaningful configuration between the two objects, the best-weighted average favored the preferred object even more. Although the interaction of task with weighting was not quite significant in Baeck et al., their results suggest that two objects in a pair are weighted differently when their configuration is attended. This raises an interesting interpretation of our expertise effects. Different parts of the same object are meaningfully related, and one theory suggests that attention to diagnostic parts is the change in expertise that can account for holistic effects (Chua et al., 2015). Even though all our subjects were asked to perform the same task, car experts may be processing cars at the subordinate level regardless of instructions, while novices may not (Gauthier et al., 1999; Tanaka & Taylor, 1991). Thus, while the results may suggest a different kind of representation in experts, they could also reflect that experts attend to all parts of an object of expertise. However, if this were the case the correlation between expertise and our configural measure ought to reverse when the top and bottom parts were weighted more or less, which we did not observe.

Limitations

This study has several limitations: One is that we only tested men, so as not to confound expertise and gender. Another is that the limited field-of-view of high-resolution imaging and the greater field inhomogeneities at 7T limited the number of face-selective ROIs (for instance we did not look at face selective areas in the anterior temporal lobe or early visual cortex). For the

same reasons, we do not address issues of connectivity among areas of the face-selective network and the rest of the brain (e.g., Wang et al., 2016).

While we had enough power to detect moderate effects of expertise, our confidence intervals on these effects are relatively large. Indeed, since we deliberately recruited subjects that varied in car expertise, our design would not be ideal to characterize the effect size in the normal population. In addition, due to the difficulty of achieving sufficient power to test differences between correlations of these magnitudes, we should be cautious to draw strong conclusions in comparing across ROIs. For similar reasons, it is not possible to make a strong claim about the multivariate effects being limited to FFA2; but it is still interesting to note that while the distinction between FFA1 and FFA2 is a relatively new trend in the literature (see Pinsk, et al., 2009; Weiner & Grill-Spector, 2010), there is already some emerging evidence that the correlation between univariate BOLD and expertise may be more prominent in FFA2 than in FFA1 (McGugin, Newton et al., 2014; McGugin, Van Gulick et al., 2014; McGugin, Gatenby et al., 2012).

Conclusions

Overall, our results provide further evidence for qualitative changes in object representations in the FFA, especially the right FFA2, as a result of expertise. In addition, they build on behavioral studies associating expertise with an increase in holistic/configural processing recognition (Gauthier & Tarr, 2002; Chua et al., 2015; Wong, Palmeri & Gauthier, 2009; Wong, Palmeri, Rogers et al., 2009; Bukach et al., 2010), demonstrating that a similar neural measure also differentiates between car experts and car novices.

References

- Avidan, G., Tanzer, M., Hadj-Bouziane, F., Liu, N., Ungerleider, L. G., & Behrmann, M. (2014). Selective dissociation between core and extended regions of the face processing network in congenital prosopagnosia. *Cerebral Cortex*, 24, 1565-1578.
- Baeck, A., Wagemans, J., & de Beeck, H. P. O. (2013). The distributed representation of random and meaningful object pairs in human occipitotemporal cortex: The weighted average as a general rule. *Neuroimage*, 70, 37-47.
- Barense, M. D., Henson, R. N., Lee, A. C., & Graham, K. S. (2010). Medial temporal lobe activity during complex discrimination of faces, objects, and scenes: Effects of viewpoint. *Hippocampus*, 20(3), 389-401.
- Bilalić, M., Grottenhaler, T., Nägele, T., & Lindig, T. (2014). The faces in radiological images: fusiform face area supports radiological expertise. *Cerebral Cortex*, 272.
- Bilalic, M., Langner, R., Ulrich, R., & Grodd, W. (2011). Many faces of expertise: Fusiform face area in chess experts and novices. *Journal of Neuroscience*, 31, 10206-20214.
- Bukach, C. M., Gauthier, I., & Tarr, M. J. (2006). Beyond faces and modularity: The power of an expertise framework. *Trends in Cognitive Sciences*, 10, 159-166.
- Bukach, C. M., Phillips, W. S., & Gauthier, I. (2010). Limits of generalization between categories and implications for theories of category specificity. *Attention, Perception, & Psychophysics*, 72, 1865-1874.
- Chang, C. C., & Lin, C. J. (2011). LIBSVM: A library for support vector machines. *ACM Transactions on Intelligent Systems and Technology (TIST)*, 2, 27.
- Chua, K.-W., Richler, J. J., & Gauthier, I. (2015). Holistic processing from learned attention to parts. *Journal of Experimental Psychology: General*, 144, 723-9.

- Davis, T., LaRocque, K. F., Mumford, J. A., Norman, K. A., Wagner, A. D., & Poldrack, R. A. (2014). What do differences between multi-voxel and univariate analysis mean? How subject-, voxel-, and trial-level variance impact fMRI analysis. *Neuroimage*, 97, 271-283.
- Duchaine, B., & Nakayama, K. (2006). The Cambridge Face Memory Test: Results for neurologically intact individuals and an investigation of its validity using inverted face stimuli and prosopagnosic participants. *Neuropsychologia*, 44, 576-585.
- Elbich, D. B., & Scherf, S. (2017). Beyond the FFA: Brain-behavior correspondences in face recognition abilities. *NeuroImage*, 147, 409-422.
- Friston, K. J., Frith, C. D., Turner, R., & Frackowiak, R. S. (1995). Characterizing evoked hemodynamics with fMRI. *Neuroimage*, 2, 157-165.
- Furl, N., Garrido, L., Dolan, R. J., Driver, J., & Duchaine, B. (2011). Fusiform gyrus face selectivity relates to individual differences in facial recognition ability. *Journal of Cognitive Neuroscience*, 23, 1723-1740.
- Gauthier, I., Curran, T., Curby, K. & Collins, D. (2003). Perceptual interference supports a non-modular account of face processing. *Nature Neuroscience*, 6, 428-432.
- Gauthier, I., Skudlarski, P., Gore, J. C., & Anderson, A. W. (2000). Expertise for cars and birds recruits brain areas involved in face recognition. *Nature Neuroscience*, 3, 191-197.
- Gauthier, I., & Tarr, M. J. (2002). Unraveling mechanisms for expert object recognition: Bridging brain activity and behavior. *Journal of Experimental Psychology: Human Perception and Performance*, 28, 431.
- Gauthier, I., Tarr, M. J., Anderson, A. W., Skudlarski, P., & Gore, J. C. (1999). Activation of the middle fusiform “face area” increases with expertise in recognizing novel objects. *Nature Neuroscience*, 2, 568-573.

- Gauthier, I., Williams, P., Tarr, M. J., & Tanaka, J. (1998). Training ‘greeble’ experts: A framework for studying expert object recognition processes. *Vision Research*, 38, 2401-2428.
- Golarai, G., Liberman, A., & Grill-Spector, K. (2017). Experience shapes the development of neural substrates of face processing in human ventral temporal cortex. *Cerebral Cortex*, 1229-1244.
- Grill-Spector, K., Kushnir, T., Hender, T. & Malach, R. (2000). The dynamics of object-selective attention correlate with recognition performance in humans, *Nature Neuroscience*, 3, 837-843.
- Grill-Spector, K., Knouf, N., & Kanwisher, N. (2004). The fusiform face area subserves face perception, not generic within-category identification. *Nature Neuroscience*, 7, 555-562.
- Harel, A., Gilaie-Dotan, S., Malach, R., Bentin, S. (2010). Top-down engagement modulates the neural expressions of visual expertise. *Cerebral Cortex*, 20, 2304–2318.
- Haist, F., Lee, K., & Stiles, J. (2010). Individuating faces and common objects produces equal responses in putative face-processing areas in the ventral occipitotemporal cortex. *Frontiers in Human Neuroscience*, 4.
- Harley, E. M., Pope, W. B., Villablanca, J. P., Mumford, J., Suh, R., Mazziotta, J. C., Enzmann, D., Engel, S. A. (2009). Engagement of fusiform cortex and disengagement of lateral occipital cortex in the acquisition of radiological expertise. *Cerebral Cortex*, 19, 2746–2754.
- Haxby, J. V., Gobbini, M. I., Furey, M. L., Ishai, A., Schouten, J. L., & Pietrini, P. (2001). Distributed and overlapping representations of faces and objects in ventral temporal cortex. *Science*, 293, 2425-2430.

- Jiang, X., Bollich, A., Cox, P., Hyder, E., James, J., Gowani, S. A., & Riesenhuber, M. (2013). A quantitative link between face discrimination deficits and neuronal selectivity for faces in autism. *NeuroImage: Clinical*, 2, 320-331.
- Kay, K.N., Weiner, K.S., Grill-Spector, K. (2015). Attention reduces spatial uncertainty in human ventral temporal cortex. *Current Biology*, 25, 595-600.
- Kravitz, D.J., Kriegeskorte, N. & Baker, C.I. (2010). High-level visual object representations are constrained by position. *Cerebral Cortex*, 20, 2916-2925.
- Kriegeskorte, N., Goebel, R., & Bandettini, P. (2006). Information-based functional brain mapping. *Proceedings of the National Academy of Sciences of the United States of America*, 103(10), 3863-3868.
- Liu, T. T., & Behrmann, M. (2014). Impaired holistic processing of left-right composite faces in congenital prosopagnosia. *Frontiers in Human Neuroscience*, 8.
- MacEvoy, S. P., & Epstein, R. A. (2011). Constructing scenes from objects in human occipitotemporal cortex. *Nature Neuroscience*, 14, 1323-1329.
- Marotta, J. J., McKeeff, T. J., & Behrmann, M. (2002). The effects of rotation and inversion on face processing in prosopagnosia. *Cognitive Neuropsychology*, 19, 31-47.
- McGugin, R. W., Gatenby, C., Gore, J. C., Gauthier, I. (2012). High-resolution imaging of expertise reveals reliable object selectivity in the FFA related to perceptual performance. *Proceedings of the National Academy of Sciences*, 109, 17063-17068.
- McGugin, R. W., Newton, A. T., Gore, J. C., Gauthier, I. (2014a). Robust expertise effects in right FFA. *Neuropsychologia*, 63, 135-144.

- 583 McGugin, R. W., Richler, J. J., Herzmann, G., Speegle, M., & Gauthier, I. (2012). The
584 Vanderbilt Expertise Test reveals domain-general and domain-specific sex effects in
585 object recognition. *Vision Research*, 69, 10-22.
- 586 McGugin, R. W., Ryan, K. F., Tamber-Rosenau, B. J. & Gauthier, I. (2017). The role of
587 experience in the face-selective response in right FFA. *Cerebral Cortex*, May 3, 1-14,
588 epub ahead of print.
- 589 McGugin, R. W., Van Gulick, A. E., Tamber-Rosenau, B. J., Ross, D. A. & Gauthier, I. (2014b).
590 Expertise effects in face selective areas are robust to clutter and diverted attention but not
591 to competition. *Cerebral Cortex*, 25, 2610-22.
- 592 Nestor, A., Plaut, D. C., & Behrmann, M. (2011). Unraveling the distributed neural code of
593 facial identity through spatiotemporal pattern analysis. *Proceedings of the National*
594 *Academy of Sciences*, 108(24), 9998-10003.
- 595 Norman, K. A., Polyn, S. M., Detre, G. J., & Haxby, J. V. (2006). Beyond mind-reading: Multi-
596 voxel pattern analysis of fMRI data. *Trends in Cognitive Sciences*, 10, 424-430.
- 597 O'Neil, E. B., Barkley, V. A., & Köhler, S. (2013). Representational demands modulate
598 involvement of perirhinal cortex in face processing. *Hippocampus*, 23(7), 592-605.
- 599 Pierce, K., Haist, F., Sedaghat, F., & Courchesne, E. (2004). The brain response to personally
600 familiar faces in autism: Findings of fusiform activity and beyond. *Brain*, 127, 2703-
601 2716.
- 602 Russell, R., Duchaine, B. & Nakayama, K. (2009). Super-recognizers: People with extraordinary
603 face recognition ability. *Psychonomic Bulletin & Review*, 16, 252-257.

- Schiltz, C., Dricot, L., Goebel, R., & Rossion, B. (2010). Holistic perception of individual faces in the right middle fusiform gyrus as evidenced by the composite face illusion. *Journal of Vision*, 10, 25.
- Schiltz, C., Sorger, B., Caldara, R., Ahmed, F., Mayer, E., Goebel, R., & Rossion, B. (2006). Impaired face discrimination in acquired prosopagnosia is associated with abnormal response to individual faces in the right middle fusiform gyrus. *Cerebral Cortex*, 16, 574-586.
- Tanaka, J. W., & Taylor, M. (1991). Object categories and expertise: Is the basic level in the eye of the beholder? *Cognitive Psychology*, 23, 457-482.
- Talairach, J., & Tournoux P. (1988). *Co-planar Stereotaxic Atlas of the Human Brain*. Thieme Medical, New York.
- Pinsk, M. A., Arcaro, M., Weiner, K. S., Kalkus, J. F., Inati, S. J., Gross, C. G., & Kastner, S. (2009). Neural representations of faces and body parts in macaque and human cortex: A comparative fMRI study. *Journal of Neurophysiology*, 101, 2581-2600.
- Pitcher, D., Walsh, V. Yovel, G. & Duchaine, B. (2007). TMS evidence for the involvement of the right occipital face area in early face processing, *Current Biology*, 17, 1568-1573.
- Ratner, K. G., Kaul, C., & Van Bavel, J. J. (2013). Is race erased? Decoding race from patterns of neural activity when skin color is not diagnostic of group boundaries. *Social Cognitive and Affective Neuroscience*, 8, 750-755.
- Van Bavel, J. J., Packer, D. J., & Cunningham, W. A. (2011). Modulation of the fusiform face area following minimal exposure to motivationally relevant faces: Evidence of in-group enhancement (not out-group disregard). *Journal of Cognitive Neuroscience*, 23, 3343-3354.

- Van Gulick, A. E., McGugin, R. W., & Gauthier, I. (2016). Measuring nonvisual knowledge about object categories: the semantic vanderbilt expertise test. *Behavior research methods*, 48(3), 1178-1196.
- Wang, X., Zhen, Z., Song, Y., Huang, L., Kong, X. & Liu, J. (2016). The hierarchical structure of the face network revealed by its functional connectivity pattern. *Journal of Neuroscience*, 36, 890-900.
- Weiner, K. S., & Grill-Spector, K. (2010). Sparsely-distributed organization of face and limb activations in human ventral temporal cortex. *Neuroimage*, 52, 1559-1573.
- Wong, A. C. N., Palmeri, T. J., & Gauthier, I. (2009). Conditions for facelike expertise with objects: Becoming a Ziggerin expert – but which type? *Psychological Science*, 20, 1108-1117.
- Wong, A. C. N., Palmeri, T. J., Rogers, B. P., Gore, J. C., & Gauthier, I. (2009). Beyond shape: How you learn about objects affects how they are represented in visual cortex. *PloS one*, 4, e8405.
- Worsley, K.J., Evans, A.C., Marrett, S., Neelin, P. A (1992). Three-dimensional statistical analysis for CBF activation studies in human brain. *Journal of Cerebral Blood Flow & Metabolism*, 12(6), 900-18.
- Xu, Y. (2005). Revisiting the role of the fusiform face area in visual expertise. *Cerebral Cortex*, 15, 1234-1242.
- Zhang, J., Liu, J., & Xu, Y. (2015). Neural decoding reveals impaired face configural processing in the right Fusiform Face Area of individuals with developmental prosopagnosia. *The Journal of Neuroscience*, 35, 1539-1548.

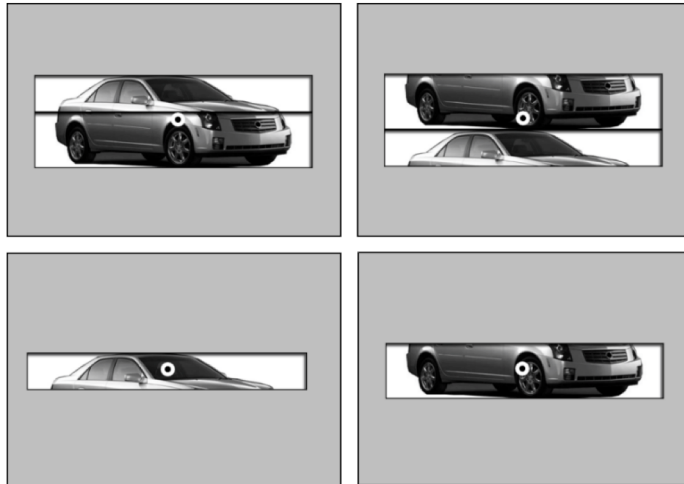


Figure 1. Example of car images used in the main experimental task. Top left: whole car; Top right: configurally-disrupted car; Bottom left: top car part; Bottom right: bottom car part.

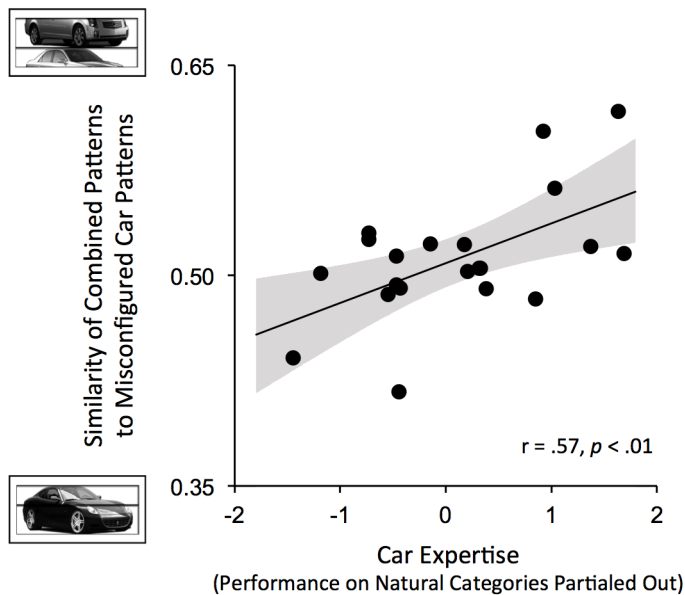


Figure 2. Similarity of combined voxel patterns to the voxel patterns elicited by configurally-disrupted cars correlated with car expertise. Combined patterns were linear combinations of the voxel patterns elicited by top and bottom car parts when presented in isolation. A probabilistic SVM classifier was trained to discriminate voxel patterns elicited by configurally-intact car images from the voxel patterns elicited by configurally-disrupted car images (top and bottom halves switched) and tested on combined patterns.

Tables

ROI	X (SD)	Y (SD)	Z (SD)	N	ROI size (SD)
Left FFA1	-39.0 (5.8)	-62.8 (8.2)	-21.0 (4.8)	20	1665 (302)
Left FFA2	-40.4 (5.7)	-45.4 (6.2)	-23.0 (5.7)	22	1668 (345)
Left OFA	-39.5 (6.6)	-81.6 (6.4)	-19.1 (4.1)	17	1513 (463)
Left LO	-45.6 (7.1)	-76.0 (8.5)	-14.3 (8.8)	24	1792 (0)
Right FFA1	38.1 (6.4)	-66.0 (9.9)	-17.1 (4.0)	21	1555 (378)
Right FFA2	38.6 (4.1)	-46.4 (5.6)	-22.6 (3.5)	21	1738 (229)
Right OFA	34.4 (5.4)	-84.0 (6.9)	-16.3 (4.3)	21	1623 (269)
Right LO	44.6 (5.5)	-75.8 (6.7)	-13.4 (7.3)	27	1792 (0)

Table 1. ROI descriptions (mean Talairach coordinates, number of subjects with a given ROI and ROI mean size, corresponding to the number of 1 mm isotropic high-res voxels).

A. Pearson correlations with car expertise

ROI	Cars - NonCars	Cars residuals
Left FFA1	.40 (.04)	.41 (.04)
Left FFA2	.33 (.06)	.34 (.06)
Left OFA	.11 (.35)	.13 (.32)
Left LO	.49 (.01)	.60 (.001)
Right FFA1	.01 (.49)	.22 (.19)
Right FFA2	.43 (.03)	.42 (.03)
Right OFA	.55 (.005)	.55 (.005)
Right LO	.36 (.03)	.32 (.05)

B. Pearson correlations with CFMT

ROI	Faces - NonFaces	Face residuals
Left FFA1	-.14 (n.a.)	-0.12 (n.a.)
Left FFA2	.34 (.06)	.31 (.08)
Left OFA	.20 (.22)	.18 (.24)
Left LO	.14 (.26)	.03 (.44)
Right FFA1	-.24 (n.a.)	-.24 (n.a.)
Right FFA2	.24 (.15)	.22 (.16)
Right OFA	.29 (.10)	.22 (.16)
Right LO	-.31 (n.a.)	-.20 (n.a.)

Table 2. Univariate expertise effects. **A.** Correlations between behavioral expertise for cars and neural selectivity for cars using a subtraction and a regression method to control for response to non-car conditions. One-tail p-values for positive relations shown in parentheses. **B.** Correlations between behavioral expertise for cars and neural selectivity for cars using a subtraction and a regression method to control for response to non-car conditions. For comparison, we show one-tail p-values for positive relations in parentheses.

	Whole cars	Misconfigured cars			Partial corr. with car exp (95% CI)
	Mean (SE)	Mean (SE)	<i>Cohen's dz</i>	<i>p</i>	
IFFA1	0.70 (0.09)	0.60 (0.09)	.27	.22	.05 (-.40, .48)
IFFA2	0.20 (0.06)	0.05 (0.07)	.50	.02	.00 (-.42, .42)
IOFA	1.21 (0.16)	1.21 (0.14)	.00	.99	.07 (-.42, .53)
ILO	0.92 (0.09)	0.92 (0.07)	.00	.97	.26 (-.16, .60)
rFFA1	1.15 (0.09)	1.10 (0.08)	.14	.52	.14 (-.31, .54)
rFFA2	0.49 (0.05)	0.36 (0.04)	.60	.01	.20 (-.25, .58)
rOFA	1.66 (0.10)	1.50 (0.07)	.47	.05	.03 (-.52, .56)
rLO	1.37 (0.07)	1.21 (0.05)	.52	.01	.43 (.06, .70)

Table 3. Main task univariate BOLD responses (with standard error), the effect size and the p-value for their difference (two-tailed paired t-test), and the correlation of the response to whole cars with car expertise, partialing out the response to misconfigured cars).

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	Whole- vs. Misconfigured Voxel Patterns Correlation with Car Expertise		Classification of Combined Voxel Patterns Classified as Whole Correlation with Car Expertise		
	Probability Correct				
	Mean (SE)	r (95%CI)	Mean (SE)	r (95%CI)	<i>p</i>
IFFA1	0.62 (0.03)	-0.02 (-.46, .43)	0.52 (0.02)	0.18 (-.29, .58)	0.54
IFFA2	0.59 (0.02)	-0.06 (-.47, .37)	0.50 (0.01)	0.04 (-.39, .45)	0.89
IOFA	0.64 (0.02)	0.29 (-.22, .68)	0.51 (0.02)	0.32 (-.19, .69)	0.23
ILO	0.60 (0.02)	0.14 (-.28, .51)	0.53 (0.01)	0.00 (-.40, .40)	0.99
rFFA1	0.63 (0.02)	0.38 (-.06, .70)	0.52 (0.02)	-0.08 (-.49, .36)	0.78
rFFA2	0.62 (0.02)	0.39 (-.05, .70)	0.49 (0.01)	0.57 (.18, .85)	0.01
rOFA	0.67 (0.02)	0.43 (.00, .73)	0.51 (0.02)	0.10 (-.35, .51)	0.73
rLO	0.62 (0.02)	0.27 (-.12, .59)	0.49 (0.01)	0.20 (-.19, .54)	0.22

681 **Table 4.** Main task classification probabilities and correlations with car expertise.